

VELTRAXX'26



Chips to Startup
Programme



TEAM NAME:

QuadraX

PROBLEM STATEMENT NO:

PS 16 (Wildcard Hardware Innovation)

PROBLEM STATEMENT TITLE:

**Hardware-Accelerated Bidirectional AES-128 Cryptographic SoC
with AXI4-Lite & Active Tamper Defense**

PS Category: Hardware (Artix-7 FPGA & Sky130 ASIC)



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PROBLEM STATEMENT

- **Real-World Problem:**

Edge IoT nodes and cyber-physical systems require high-throughput bidirectional data encryption/decryption, standard bus attachment, and active physical tamper defense under severe power and area limits.

- **Why It Matters:**

Software AES on embedded MCUs consumes 1,136+ CPU cycles/block, lacks physical fault containment, and leaves cryptographic keys vulnerable to power glitch and laser fault-injection attacks.

- **Current Challenges:**

Area explosion from separate forward/inverse S-Box ROMs, lack of memory-mapped bus interfaces in isolated IP blocks, and failure to immediately wipe out secret keys during active hardware tampering.

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YOUR SOLUTION

- **Project Name:**

QuadraX: Production-Grade Bidirectional AES-128 Crypto Accelerator SoC

- **One-Line Solution:**

A fully integrated 32-bit AMBA AXI4-Lite cryptographic SoC featuring a unified composite Galois Field $GF(2^8)$ S-Box/InvS-Box core and active 1-cycle key zeroization under tamper attack.

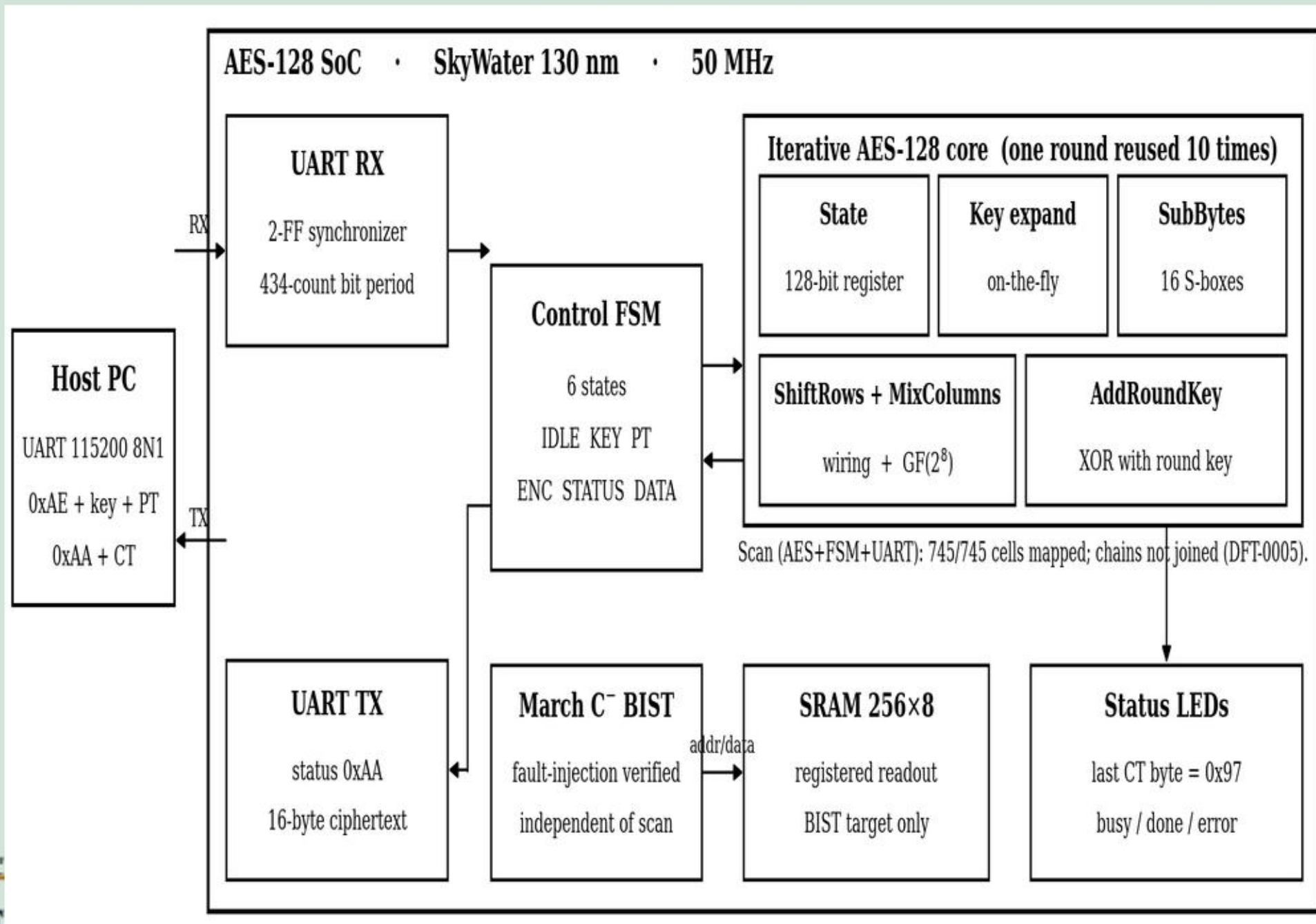
- **How It Solves the Problem:**

Shares 100% of the core inversion logic between encryption and decryption (zero table duplication), exposes 32-bit AXI registers, and instantly wipes all 11 round keys within 1 clock cycle upon fault trigger.

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ARCHITECTURAL BLOCK DIAGRAM



Enhanced SoC Architecture Highlights:

- 32-bit AMBA AXI4-Lite Slave Interface (Registers 0x00 - 0x34)
- Unified Composite GF(2⁸) Inversion (Shared SubBytes/InvSubBytes)
- Bidirectional Datapath (10-cycle Encrypt & Decrypt @ 50 MHz)
- Active Parity Monitor & 1-Cycle Key Zeroization (Atomic 0x00 Wipe)
- Unmaskable Hardware Security Interrupt (security_irq output)
- Fully Synthesizable in Yosys / OpenLane 2 (Sky130) & Vivado (Basys3)

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PURPOSE OF BLOCKS & ARCHITECTURAL TRADE-OFFS

- F**
- **Input Block (AMBA AXI4-Lite Slave Register Interface):**
Exposes 32-bit memory-mapped registers for Control (0x00), Status (0x04), Key (0x08-0x14), and Data In (0x18-0x24) with non-blocking handshakes.
 - **Processing Block (Bidirectional AES Core + Composite GF S-Box + Parity Monitor):**
Executes 10-cycle AES-128 Encryption and Decryption using unified Galois Field inversion; continuously monitors state parity across round steps.
 - **Output & Security Block (DOUT Registers + 1-Cycle Zeroization Engine + IRQ):**
Outputs 128-bit results at 0x28-0x34; instantly wipes all 11 round keys to 0x00 and fires security_irq within 1 clock cycle upon fault trigger.

Architectural Trade-Off Decisions:

- **Unified Composite GF(2⁸) Sharing:**
Shares 100% of multiplicative inversion between Enc & Dec; saves over 40% area vs instantiating duplicate inverse S-Box ROMs.
- **Active 1-Cycle Hardware Zeroization:**
Sacrifices ongoing transaction to guarantee complete protection against differential fault analysis (DFA) key recovery attacks.

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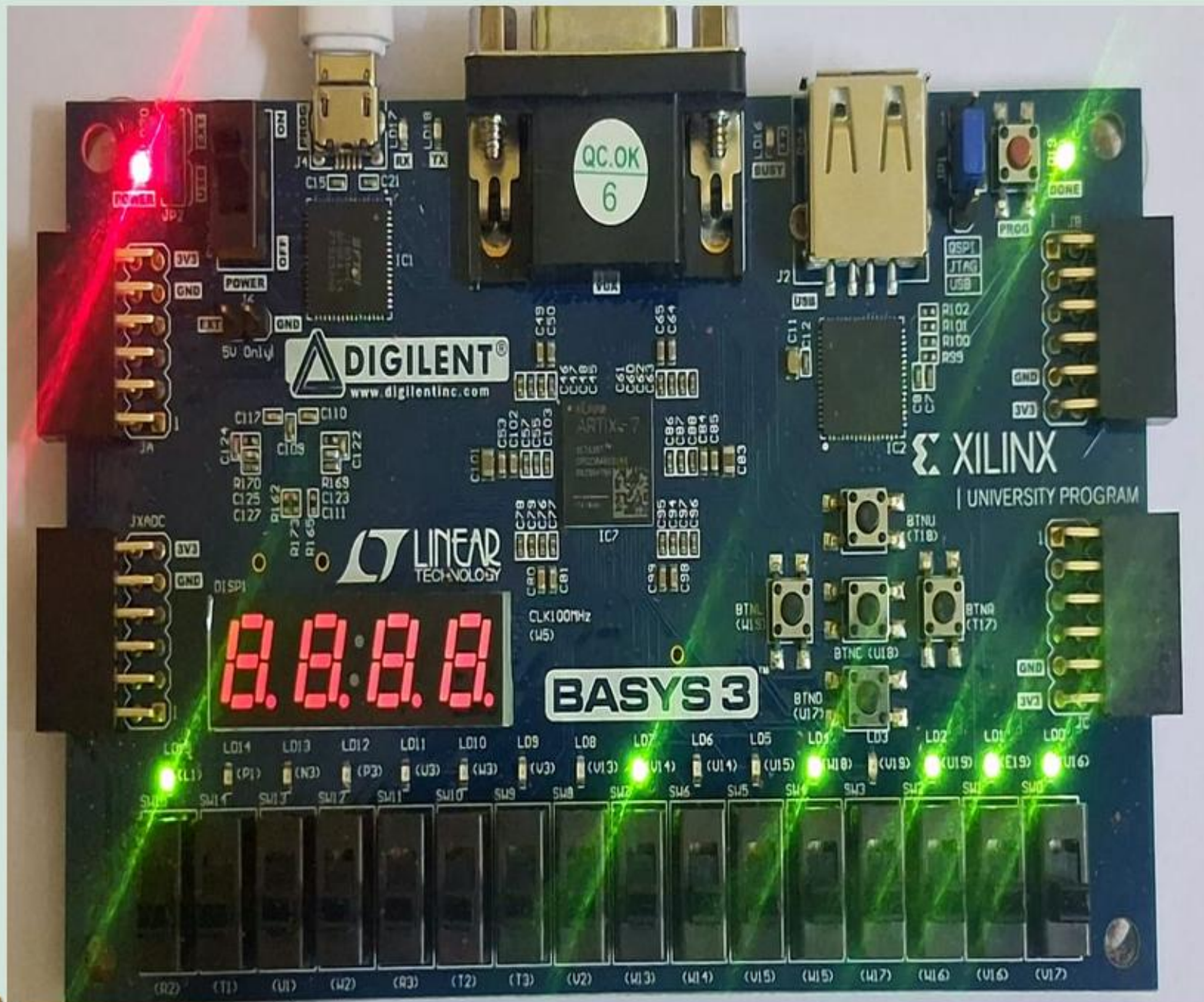
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OUTPUTS AND RESULTS: VERIFICATION & HARDWARE DEMO



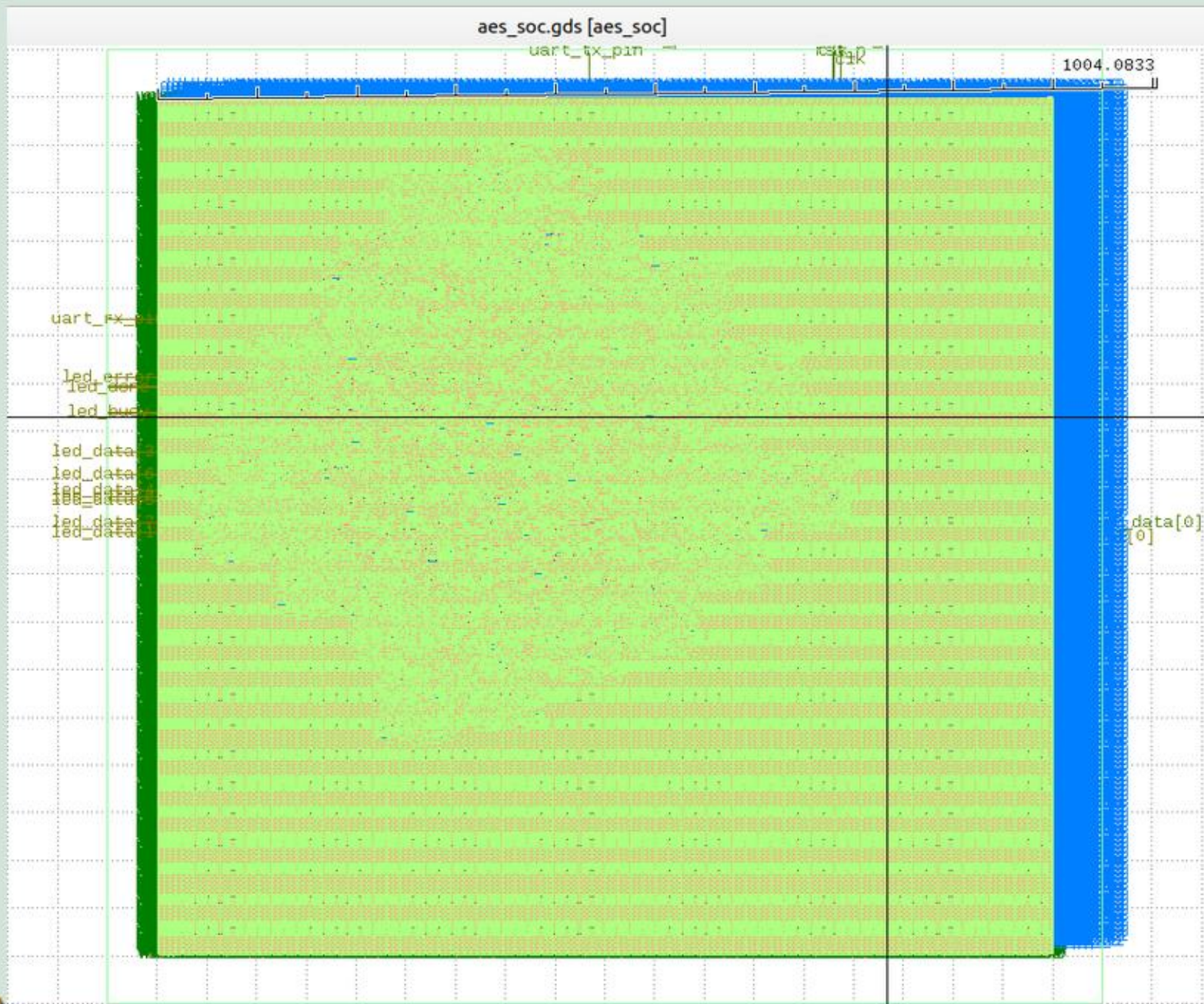
Measured Verification & Hardware Results:

- **Target Platform:**
Digilent Basys3 Artix-7 FPGA (xc7a35tcp236-1) & Sky130 ASIC
- **Clock Frequency:**
50 MHz Operating Clock (20.0 ns Period)
- **Latency & Throughput:**
10 Clock Cycles = 200 ns per 128-bit block (0.64 Gbps sustained)
- **NIST SP 800-38A Encryption KAT:**
PASS: CT = 3ad77bb4 0d7a3660 a89ecaf3 2466ef97 (100% exact match)
- **NIST SP 800-38A Decryption KAT:**
PASS: Recovered PT = 6bc1bee2 2e409f96 e93d7e11 7393172a (100% match)
- **1-Cycle Key Zeroization Test:**
PASS: 1-cycle core abort, security_irq asserted, all keys wiped to 0x00

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OUTPUTS AND RESULTS: ASIC SIGNOFF & SYNTHESIS



Physical Signoff & Synthesis Metrics:

- **PDK & Implementation Flow:**
SkyWater 130nm (sky130_fd_sc_hd) via OpenLane 2 / OpenROAD
- **RTL Synthesis Signoff:**
100% Synthesizable in Yosys v0.52 (Clean Netlist in outputs/aes_soc_netlist.v)
- **STA Timing Signoff:**
0 violations across all 9 PVT corners (+7.22 ns worst setup slack)
- **Physical Verification:**
DRC: Clean (0 errors Magic/KLayout) | LVS: Clean Match (Netgen)
- **Power Analysis:**
6.08 mW typical power @ 50 MHz | Worst-case IR drop: 0.021%
- **Design-for-Test (DFT):**
745/745 scan cells mapped (100% sequential coverage) + MBIST verified



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RESEARCH AND REFERENCES

References & Technical Artifacts:

- 1. NIST FIPS PUB 197: Advanced Encryption Standard (AES) Specification, National Institute of Standards and Technology, Nov. 2001.
- 2. NIST SP 800-38A: Recommendation for Block Cipher Modes of Operation — Methods and Techniques (ECB Known-Answer Test Vectors).
- 3. ARM AMBA AXI4-Lite Protocol Specification, ARM Limited, Cambridge, UK.
- 4. SkyWater 130nm Open-Source PDK (sky130_fd_sc_hd) & OpenLane 2 / OpenROAD ASIC Implementation Flow.
- 5. Team QuadraX Hackathon GitHub Repository: <https://github.com/akifmohamed/VELTRAXX26-PS16-QuadraX>